

Mobile identity-verification plugin

1. What this component is, and what it is not

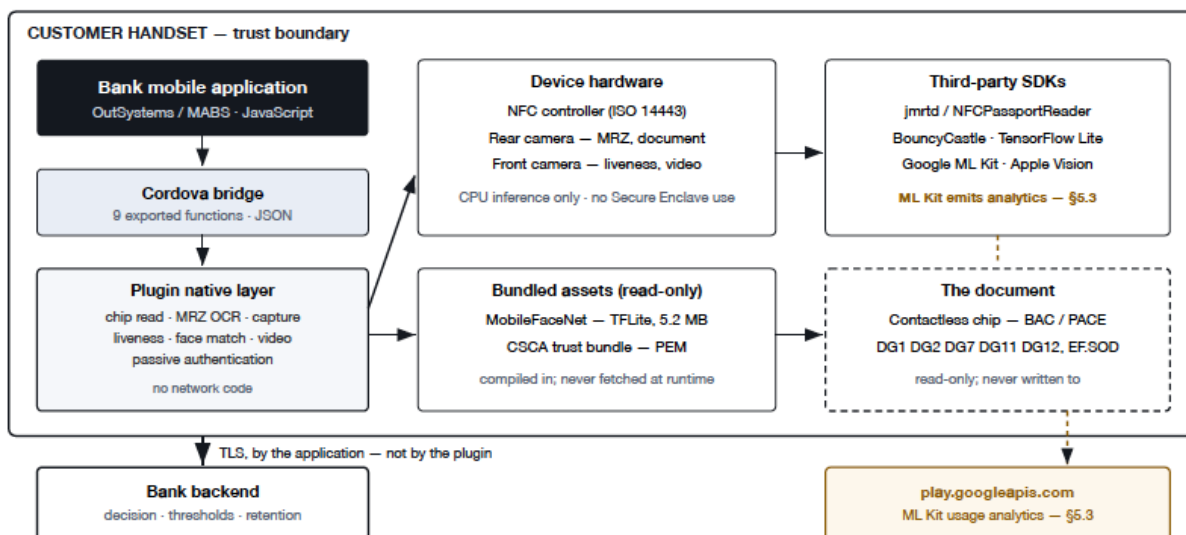
A Cordova plugin embedded in the bank's mobile application. It reads ICAO 9303 electronic identity documents (passports and national ID cards) over NFC, scans the printed machine-readable zone with the camera, photographs the document, runs a challenge–response liveness check, and compares the holder's face against the portrait stored on the document's chip.

It is a data-collection and evidence-gathering component. It is not a decision engine.

It returns measurements — a chip payload, a similarity score, a set of checks that passed or failed — to the host application, which forwards them to the bank's backend. The accept/reject decision, and every threshold behind it, lives in the backend and is deliberately not shipped in the app.

Property	Value
Execution context	In-process native code inside the bank's own mobile application
Processing location	Entirely on the customer's handset
Server-side components	None. The plugin has no backend of its own.
Outbound network connections made by the plugin	None
Data persisted by the plugin	None beyond transient temporary files, deleted within the same call
Platforms	Android (Java), iOS 15.5+ (Swift 5)

2. Component architecture



3. Third-party framework inventory

I. Android

Library	Version	Licence	Function
org.jmrtd:jmrtd	0.7.42	LGPL 3.0	ICAO 9303 protocol: BAC/PACE, EF.SOD, data groups
net.sf.scuba:scuba-sc-android	0.0.26	LGPL 3.0	Smart-card transport beneath jmrtd
org.bouncycastle:bcprov-jdk18on	1.78	MIT-style	Cryptography for passive authentication
edu.ucar:jj2000	5.2	BSD-style	JPEG 2000 decoding of the DG2 portrait
com.google.mlkit:text-recognition	16.0.0	Google ToS	MRZ OCR, document evidence check
com.google.mlkit:face-detection	16.1.7	Google ToS	Liveness challenges, face cropping
androidx.camera:*	1.3.1	Apache 2.0	Camera preview, frame analysis, video recording
androidx.appcompat:appcompat	1.6.1	Apache 2.0	Activity theming
org.tensorflow:tensorflow-lite	2.14.0	Apache 2.0	Face-match model inference

II. iOS

Pod	Resolved	Licence	Function
NFCPassportReader	1.1.9	MIT	ICAO 9303 over CoreNFC
GoogleMLKit/FaceDetection	8.0.0	Google ToS	Liveness challenges, face cropping
TensorFlowLiteSwift	2.14.x	Apache 2.0	Face-match model inference

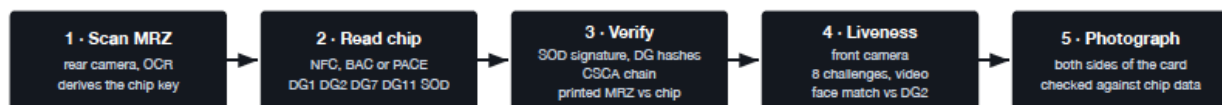
Apple frameworks used directly, with no third party involved:

CoreNFC, Vision (all OCR on iOS), AVFoundation, UIKit, CoreGraphics, ImageIO. OCR on iOS is Apple Vision, not ML Kit — and Vision recognises Arabic, which MLKit has no model for.

III. Bundled model

Property	Value
Model	MobileFaceNet, TensorFlow Lite, 5.2 MB, compiled into the app
Input / output	112 x 112 RGB → 192-dimensional embedding
Comparison	Cosine similarity, computed on-device
Decision threshold	Not present in the plugin. The similarity is returned; the backend decides.
Provenance	<code>src/models/PROVENANCE.md</code> — model governance sign-off outstanding

4. Processing flow



Why this order

The MRZ must be read first because it derives the chip access key. The chip must be read before the photographs, so the print-versus-chip comparison happens while the customer still holds the document.

Single JSON result → host application → bank backend over the application's own TLS channel

document fields · DG2 portrait · passive-authentication verdict · MRZ/chip comparison · liveness result

challenge frames · session video · face-match similarity (a number, not a decision) · document photograph

Nothing is written to durable storage by the plugin. Temporary video files are deleted within the same call.